

The Editors
Social Networks
 Elsevier

Dear Editors,

We submit for your consideration our manuscript **“The Camouflage Ratio: Strategic Tie Formation and the Limits of Structural Detection”** as a research article.

The paper concerns one of the oldest uses of relational data: inferring an actor’s attributes from those of its alters. That inference rests on homophily, and the journal has published extensively both on measuring homophily and on the conditions under which neighbour-based inference is reliable. We ask what happens when an actor *chooses* its ties in order to defeat the inference—and we show that the answer is not simply “the inference degrades”.

Three results may interest your readership. First, camouflage is not a private act. Because a tie has two ends, an actor that forms ties outside its own category necessarily contaminates its alters’ neighbourhoods in turn, and edge-endpoint conservation fixes the rate at which it does so. The consequence is counterintuitive: prevalence acts as protection, so the commoner such actors are, the less camouflage each one needs. Second, we derive in closed form the point at which averaging over alters ceases to improve on attributes alone, and show that selectively discarding suspicious ties does not straightforwardly repair it—discarding trades bias against the number of ties left to average over, so its benefit is bounded by the quality of the suspicion score rather than by the discarding itself. Third, we measure the quantity on a population of 10,199 labelled Twitter accounts and find that every recorded relation is one on which the inference is harmed, while the same accounts behave in opposite ways across relation layers: they avoid one another where ties are visible and converge where content is shared. The measurement is therefore a per-relation quantity, not a property of a population—a point on which we build directly on the journal’s multiplexity literature.

We would like to be explicit about priority, since it shapes what we claim. That a crossover of this kind exists is not our discovery: for balanced classes it is a result of Ma et al. (2022), and the bounded interval above which relational information becomes informative again is the second root of the same condition, discussed by Luan et al. (2023) as the mid-homophily pitfall. Our contribution is the asymmetric, class-imbalanced case, together with the empirical measurement. We state this in the abstract, the introduction and the conclusion rather than leaving a reader to discover it.

We also note two things a referee may wish to probe, and which we have tried not to obscure. The attribute-separation parameter on which our empirical thresholds depend is calibrated from the achievable error of a held-out linear discriminant rather than from the moments of the attribute distributions, because the moments-based estimate is badly inflated on high-dimensional, anisotropic attributes; we report the calibration and its consequences. And our identity is exact in the model but requires a degree correction on data whose classes differ in mean degree, which we document.

The manuscript is original, is not under consideration elsewhere, and has not been published previously. All authors have approved the submission and declare no competing interests. In accordance with the journal’s double-anonymized review policy the manuscript itself carries no author information; author details and the CRediT statement are in the separate title page. All data analysed are third-party and publicly available, as set out in the data availability statement; we also supply the code needed to reproduce every reported number.

Thank you for considering our work.

Yours sincerely,

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